

2ai	Constant amplitude	B1
2aii	Period = 0.75 s $\omega = 2\pi / T$ $\omega = 8.4 \text{ rad s}^{-1}$	C1 A1

2aiii	Either use of gradient or $v = \omega y_0$ $v = 0.168 \text{ m s}^{-1}$ (allow ± 0.02 for construction)	C1 A1
2bi	$f = 1/T = 1/0.75$ $= 1.3 \text{ Hz}$	B1
2bii	at $\frac{1}{2} f_0$ 'pulse' / energy is provided to mass on <u>alternate/some</u> oscillations so 'pulses' / energy <u>build up</u> the amplitude (don't accept maximum amplitude)	M1 A1
2bi	Transverse	A1